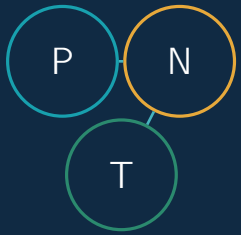
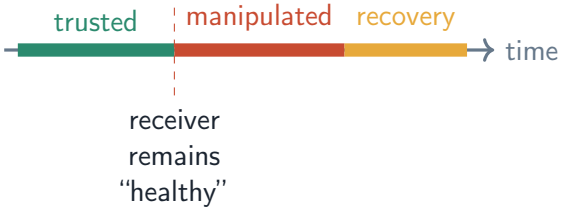
Assured PNT

Navigation When GPS Cannot Be Trusted

Michael L. Carroll
Kalman Filter Theory and Applications



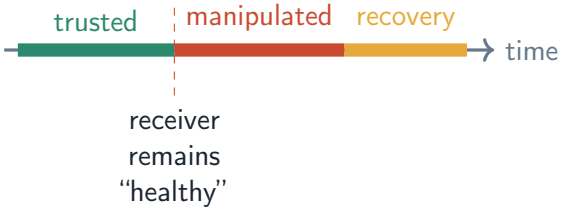
A confident answer can still be wrong



The dangerous case

The navigation computer continues to report a precise solution while the external measurement has become misleading.

A confident answer can still be wrong



- Availability is not integrity.
- Precision is not truth.
- Fusion is not automatic trust.

The dangerous case

The navigation computer continues to report a precise solution while the external measurement has become misleading.

Source: FAA, *GNSS Interference Resource Guide*, 2025

What PNT provides



What PNT provides



What PNT provides

P

Position

Where am I?

N

Navigation

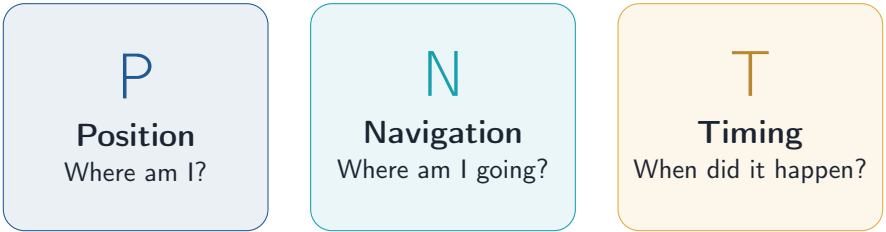
Where am I going?

T

Timing

When did it happen?

What PNT provides



A resilient system must preserve the **mission-relevant combination** of position, velocity, attitude, and time.

GPS is powerful—and physically fragile

Why we use it

- Global, common reference
- Position, velocity, and precise time
- Excellent long-term stability
- Low user-equipment burden

GPS is powerful—and physically fragile

Why we use it

- Global, common reference
- Position, velocity, and precise time
- Excellent long-term stability
- Low user-equipment burden

Why it needs help

- Very low received signal power
- Blockage, multipath, and interference
- Jamming can remove measurements
- Spoofing can create plausible false data

Source: GPS.gov resilience guidance; FAA GNSS Interference Resource Guide

GPS is powerful—and physically fragile

Why we use it

- Global, common reference
- Position, velocity, and precise time
- Excellent long-term stability
- Low user-equipment burden

Why it needs help

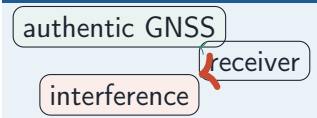
- Very low received signal power
- Blockage, multipath, and interference
- Jamming can remove measurements
- Spoofing can create plausible false data

Design for **loss** and **manipulation**, not merely nominal accuracy.

Source: GPS.gov resilience guidance; FAA GNSS Interference Resource Guide

Two failures, two different hazards

Jamming: deny the signal



- The authentic signal is overwhelmed.
- Loss or degradation may be obvious.

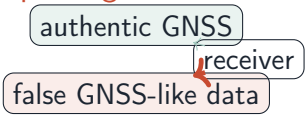
Two failures, two different hazards

Jamming: deny the signal



- The authentic signal is overwhelmed.
- Loss or degradation may be obvious.

Spoofing: counterfeit the signal



- The false signal looks plausible.
- Bad data can contaminate fused systems.

Source: FAA, *GNSS Interference Resource Guide*, Sections 1.5–1.7

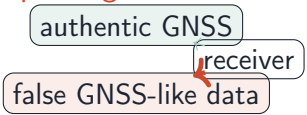
Two failures, two different hazards

Jamming: deny the signal



- The authentic signal is overwhelmed.
- Loss or degradation may be obvious.

Spoofing: counterfeit the signal



- The false signal looks plausible.
- Bad data can contaminate fused systems.

A Kalman filter can reject inconsistent data; it does not, by itself, authenticate GNSS.

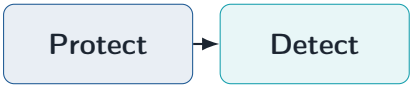
Source: FAA, *GNSS Interference Resource Guide*, Sections 1.5–1.7

Assurance is a behavior, not a sensor

Protect

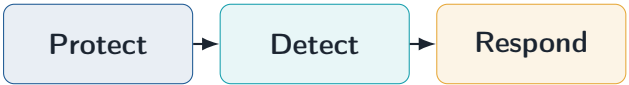
Source: NIST IR 8323 Rev. 1; DHS Resilient PNT Reference Architecture

Assurance is a behavior, not a sensor



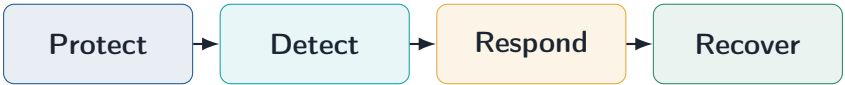
Source: NIST IR 8323 Rev. 1; DHS Resilient PNT Reference Architecture

Assurance is a behavior, not a sensor



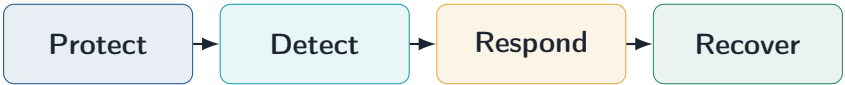
Source: NIST IR 8323 Rev. 1; DHS Resilient PNT Reference Architecture

Assurance is a behavior, not a sensor



Source: NIST IR 8323 Rev. 1; DHS Resilient PNT Reference Architecture

Assurance is a behavior, not a sensor



Protect	Detect	Respond	Recover
Prevent avoidable exposure	Recognize anomalies quickly	Continue at known performance	Re-admit sources safely

Source: NIST IR 8323 Rev. 1; DHS Resilient PNT Reference Architecture

Assurance is a behavior, not a sensor



Protect	Detect	Respond	Recover
Prevent avoidable exposure	Recognize anomalies quickly	Continue at known performance	Re-admit sources safely

The system must know what it trusts—and why.

Source: NIST IR 8323 Rev. 1; DHS Resilient PNT Reference Architecture

Inertial navigation provides the bridge

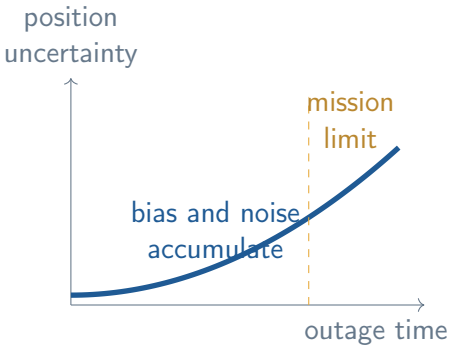
INS strengths

- Self-contained after initialization
- High-rate position, velocity, and attitude
- Immune to radio-frequency jamming
- Smooth short-term propagation

Inertial navigation provides the bridge

INS strengths

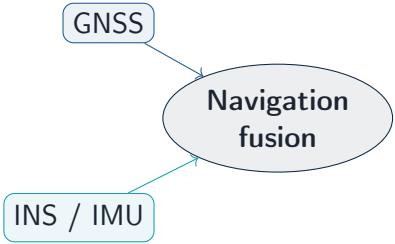
- Self-contained after initialization
- High-rate position, velocity, and attitude
- Immune to radio-frequency jamming
- Smooth short-term propagation



INS limitation

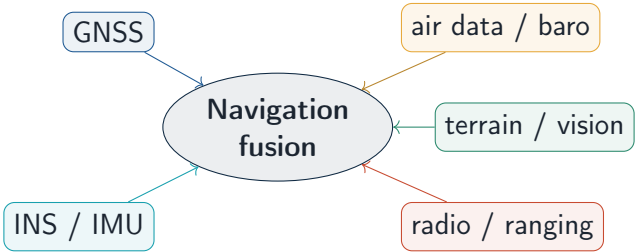
Without trusted aiding, uncertainty grows and the solution eventually exceeds the mission's error bound.

Resilience comes from complementary errors



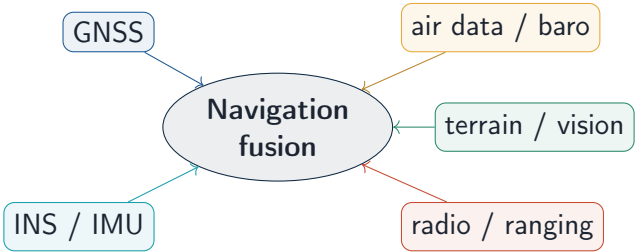
Source: U.S. DOT Complementary PNT Action Plan, 2024

Resilience comes from complementary errors



Source: U.S. DOT Complementary PNT Action Plan, 2024

Resilience comes from complementary errors

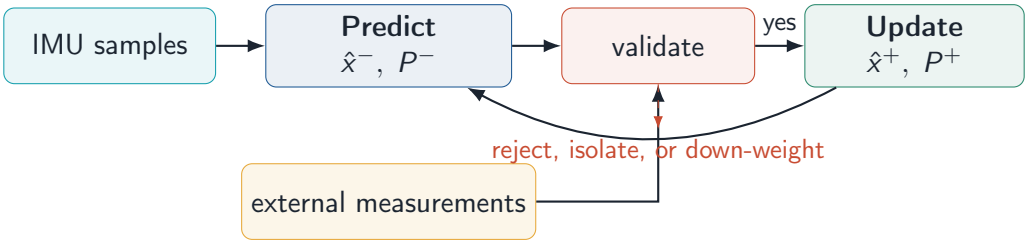


The design test

Sources are valuable when their failure modes are sufficiently independent and their measurements make the needed states observable.

Source: U.S. DOT Complementary PNT Action Plan, 2024

The Kalman filter is the integration engine



Prediction carries the solution through gaps. **Updates** bound drift—only when they are trustworthy.

A practical error-state model

$$\delta x = \begin{bmatrix} \delta r & \delta v & \delta \theta & b_a & b_g & b_c & d_c \end{bmatrix}^T$$

A practical error-state model

$$\delta x = \begin{bmatrix} \delta r & \delta v & \delta\theta & b_a & b_g & b_c & d_c \end{bmatrix}^T$$

Group	Meaning	Why estimate it?
$\delta r, \delta v$	position and velocity errors	navigation output
$\delta\theta$	small attitude errors	correct force projection
b_a, b_g	accelerometer and gyro biases	control inertial drift
b_c, d_c	clock bias and drift	align measurements in time

A practical error-state model

$$\delta x = \begin{bmatrix} \delta r & \delta v & \delta\theta & b_a & b_g & b_c & d_c \end{bmatrix}^T$$

Group	Meaning	Why estimate it?
$\delta r, \delta v$	position and velocity errors	navigation output
$\delta\theta$	small attitude errors	correct force projection
b_a, b_g	accelerometer and gyro biases	control inertial drift
b_c, d_c	clock bias and drift	align measurements in time

The state is a design choice: estimate what the mission must control.

Trust begins with the innovation

$$\nu_k = z_k - h\left(\hat{x}_k^-\right)$$

The innovation asks: *What did the sensor say that the predicted state did not expect?*

Trust begins with the innovation

$$\nu_k = z_k - h\left(\hat{x}_k^-\right)$$

The innovation asks: *What did the sensor say that the predicted state did not expect?*

$$\mathrm{NIS}_k = \nu_k^T S_k^{-1} \nu_k$$

Trust begins with the innovation

$$\nu_k = z_k - h\left(\hat{x}_k^-\right)$$

The innovation asks: *What did the sensor say that the predicted state did not expect?*

$$\mathrm{NIS}_k = \nu_k^T S_k^{-1} \nu_k$$

Before accepting an update

- Check statistical consistency.
- Compare independent sources.
- Monitor signal and receiver health.
- Apply persistence and recovery logic.

Trust begins with the innovation

$$\nu_k = z_k - h\left(\hat{x}_k^-\right)$$

The innovation asks: *What did the sensor say that the predicted state did not expect?*

$$\mathrm{NIS}_k = \nu_k^T S_k^{-1} \nu_k$$

Before accepting an update

- Check statistical consistency.
- Compare independent sources.
- Monitor signal and receiver health.
- Apply persistence and recovery logic.

A residual gate is evidence of inconsistency—not proof of the cause.

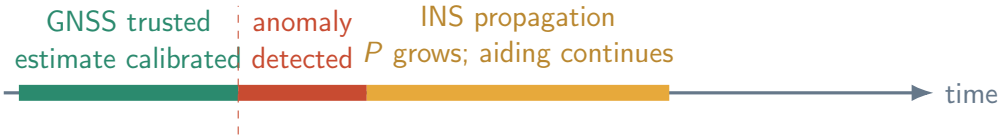
What the system should do during an outage



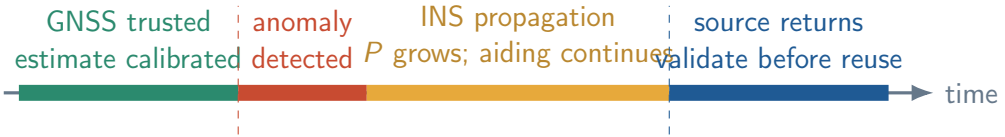
What the system should do during an outage



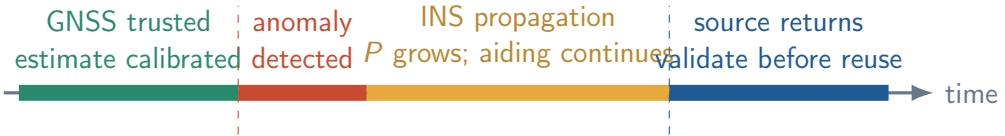
What the system should do during an outage



What the system should do during an outage



What the system should do during an outage



Graceful degradation

Continue the mission only while the estimated uncertainty remains inside a declared operational limit; otherwise transition to a safe mode.

Four design rules for assured PNT

1. **Diversify**: combine sources with different physics and failure modes.

Source: DHS Resilient PNT Reference Architecture; DHS RPNT Best Practices, 2025

Four design rules for assured PNT

1. **Diversify**: combine sources with different physics and failure modes.
2. **Manage trust**: never equate a precise measurement with an authentic one.

Source: DHS Resilient PNT Reference Architecture; DHS RPNT Best Practices, 2025

Four design rules for assured PNT

1. **Diversify**: combine sources with different physics and failure modes.
2. **Manage trust**: never equate a precise measurement with an authentic one.
3. **Expose uncertainty**: propagate covariance and compare it with mission limits.

Source: DHS Resilient PNT Reference Architecture; DHS RPNT Best Practices, 2025

Four design rules for assured PNT

1. **Diversify**: combine sources with different physics and failure modes.
2. **Manage trust**: never equate a precise measurement with an authentic one.
3. **Expose uncertainty**: propagate covariance and compare it with mission limits.
4. **Plan recovery**: define how sources are isolated, tested, and safely re-admitted.

Source: DHS Resilient PNT Reference Architecture; DHS RPNT Best Practices, 2025

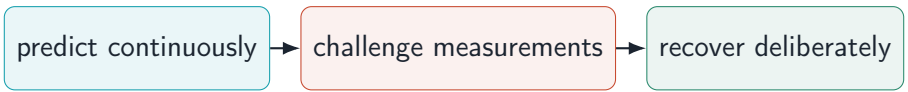
Four design rules for assured PNT

1. **Diversify**: combine sources with different physics and failure modes.
2. **Manage trust**: never equate a precise measurement with an authentic one.
3. **Expose uncertainty**: propagate covariance and compare it with mission limits.
4. **Plan recovery**: define how sources are isolated, tested, and safely re-admitted.

Assurance is the controlled behavior of the whole system.

Source: DHS Resilient PNT Reference Architecture; DHS RPNT Best Practices, 2025

The central idea



A resilient navigator does not promise perfect availability.

It promises known performance, visible uncertainty, and safe behavior.

Next: alternative aiding sources and terrain-referenced navigation.

Primary references

- Federal Aviation Administration, *GPS and GNSS Interference Resource Guide*, 2025. https://www.faa.gov/about/office_org/headquarters_offices/avs/offices/afx/afs400/afs410/GNSS
- GPS.gov, *Resilience Through Responsible Use of PNT*. https://www.gps.gov/resilience-through-responsible-use-pnt
- NIST IR 8323 Rev. 1, *Foundational PNT Profile*, 2023. https://doi.org/10.6028/NIST.IR.8323r1
- DHS, *Resilient PNT Reference Architecture*, 2022. https://www.dhs.gov/sites/default/files/2022-06/22_0609_st_resilient_pnt_ra.pdf
- U.S. DOT, *Complementary PNT Action Plan*, 2024. https://www.transportation.gov/pnt/complementary-pnt-action-plan
- DHS, *Best Practices for Resilient PNT Supporting Critical Infrastructure*, 2025.